Part 1: Linear Algebra

Q1: $AB = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

$= \begin{bmatrix} 3\times1+4\times0 & 3\times2+4\times(-1) \\ 2\times1+1\times0 & 2\times2+1\times(-1) \end{bmatrix}$

$= \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$

$BA = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1\times3+2\times2 & 1\times4+2\times1 \\ 0\times3+(-1)\times2 & 0\times4+(-1)\times1 \end{bmatrix}$

$= \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$

$(A+B)^T$: $A+B = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}$

$(A+B)^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$

AB^T: $B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$

$AB^T = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$

$= \begin{bmatrix} 3\times1+4\times2 & 3\times0+4\times(-1) \\ 2\times1+1\times2 & 2\times0+1\times(-1) \end{bmatrix}$

$= \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$

trace(A) = 3+1 = 4

$AB = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ trace(AB) = 3+3 = 6

Q2: $\det(M-\lambda I) = 0$

$M-\lambda I = \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$

$\det(M-\lambda I) = (2-\lambda)^2 - (-1)\times(-1)$

$= (2-\lambda)^2 - 1$

$= \lambda^2 - 4\lambda + 3$

$\lambda^2 - 4\lambda + 3 = 0$

$(\lambda-1)(\lambda-3) = 0$

$\lambda_1 = 1$, $\lambda_2 = 3$

when $\lambda=1$: $(M-I)v = 0$

$M - I = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$v = \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x - y = 0 \rightarrow x = y$

① $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

when $\lambda=3$: $(M-3I)\cdot v = 0$ $M-3I = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$

$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow -x - y = 0 \rightarrow x = -y$

② $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

when $\lambda_1 = 1$ $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$Mv_1 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\times1 - 1\times1 \\ -1\times1 + 2\times1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda v_1 = 1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $Mv_1 = \lambda_1 v_1$

when $\lambda_2 = 3$; $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$Mv_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2\times1 + (-1)(-1) \\ -1\times1 + 2\times(-1) \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$

$\lambda_2 \cdot v_2 = 3 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$ $Mv_2 = \lambda_2 v_2$

Answer: $\lambda_1 = 1$, $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\lambda_2 = 3$ $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Q3: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$

$A \cdot B = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$

$\therefore \text{trace}(AB) = ae + bg + cf + dh.$

$B \cdot A = \begin{bmatrix} ea+fc & eb+fd \\ ga+hc & gb+hd \end{bmatrix}$

$\therefore \text{trace}(BA) = ea + fc + gb + hd$

$\therefore \text{trace}(AB) = \text{trace}(BA)$

For general matrices: $\text{tr}(AB) = \sum_i (AB)_{ij} = \sum_i \sum_j a_{ij} b_{ji}$

$= \sum_j \sum_i b_{ji} a_{ij} = \text{tr}(BA)$

Q4: we need to prove $\det(C) \neq 0$

$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$ $\therefore \det(C) = 4 \times 6 - 7 \times 2$

$= 24 - 14 = 10 \neq 0$

$\therefore C^{-1}$ exists

for $C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $C^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$a = 4 \quad b = 7 \quad c = 2 \quad d = 6$

$\therefore C^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$

$C^{-1} = \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \cdot \frac{1}{10}$

$C \cdot C^{-1} = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} \cdot \frac{1}{10}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$

Part 2:

Q1: (a) $\sum_{k=0}^{\infty} P(X=k) = 1$

$\sum_{k=0}^{\infty} c \cdot \frac{3^k}{k!} = 1 \Rightarrow c \cdot \sum_{k=0}^{\infty} \frac{3^k}{k!} = 1$

$\because e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \therefore \sum_{k=0}^{\infty} \frac{3^k}{k!} = e^3$

$\therefore ce^3 = 1 \quad c = e^{-3}$

$P(X=k) = e^{-3} \cdot \frac{3^k}{k!}$ ∴

↑ Poisson ∴ $E[X] = 3 \quad Var(X) = 3$

Q2: $P(A) = 0.5 \quad P(B) = 0.3 \quad P(A|B) = 0.6$

(a, $P(A \cap B)$): $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$\therefore P(A \cap B) = P(A|B) \cdot P(B)$

$= 0.6 \times 0.3 = 0.18$

(b, $P(B|A)$): $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.5} = 0.36$

(c, $P(A \cup B)$): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.3 - 0.18 = 0.62$

Q3: H H T H T Head => T=2

$P(H)=\theta$ $P(T)=1-\theta$

(a) Likelihood: $L(\theta)=\theta^3(1-\theta)^2$

$l(\theta)=\log L(\theta)=3\log\theta+2\log(1-\theta)$

$\frac{dl}{d\theta}=\frac{3}{\theta}-\frac{2}{1-\theta}$

when $\frac{3}{\theta}-\frac{2}{1-\theta}=0 \rightarrow \theta=\frac{3}{5}=0.6$

(b) MAP: $P(\theta)=2\theta$, $0\le\theta\le1$ $D=\{H,H,T,H,T\}$

$P(\theta|D) \propto P(D|\theta)P(\theta)$

$P(D|\theta)=\theta^3(1-\theta)^2$ $P(\theta)=2\theta$

$\theta^4(1-\theta)^2$

$l(\theta)=4\log\theta+2\log(1-\theta)$

$\frac{dl}{d\theta}=\frac{4}{\theta}-\frac{2}{1-\theta} \rightarrow \theta=\frac{2}{3}\approx0.66$

Q4:

$$\mu = \frac{1+3+5+7}{4} = \frac{16}{4} = 4$$

$$\sigma^2 = \frac{1}{4}[(1-4)^2+(3-4)^2+(5-4)^2+(7-4)^2]$$

$$= \frac{20}{4} = 5$$

Q5: $P(\mu|x) \propto P(x|\mu)P(\mu)$

$x=5$. Likelihood: $x|\mu \sim N(\mu, 4)$

$$P(x|\mu)P(\mu) \propto e^{-\frac{(5-\mu)^2}{8}} \cdot e^{-\frac{\mu^2}{2}}$$

$$\frac{d \log p(\mu|x)}{d\mu} = \frac{5-\mu}{4} - \mu = 0$$

$$5 - \mu - 4\mu = 0$$

$$\mu = 1$$

Q6: $Var(X) = E[(X - E[X])^2]$

If $\mu = E[X] \rightarrow Var(X) = E[(X-\mu)^2]$

$$= E[X^2 - 2X\mu + \mu^2]$$

$$= E[X^2] - 2\mu E[X] + E[\mu^2]$$

for μ is constant: $E[\mu^2] = \mu^2$

$$\therefore Var(X) = E[X^2] - 2\mu \cdot \mu + \mu^2$$

$$= E[X^2] - \mu^2 = E[X^2] - [E(X)]^2$$

Part 3.

Q1: Let $Y = u^T X$ $\therefore Var(Y) = E[(Y - E[Y])^2]$

Since: $E[Y] = E[u^T X] = u^T E[X]$

Let: $\mu = E[X]$ Then: $E[Y] = u^T \cdot \mu$

$$\begin{aligned} Var(Y) = Var(u^T X) &= E[(u^T X - u^T \mu)^2] \\ &= E[(u^T (X - \mu))^2] \\ &= E[u^T (X - \mu) \cdot (u^T (X - \mu))^T] \\ &= E[u^T (X - \mu)(X - \mu)^T \cdot u] \\ &= u^T \cdot E[(X - \mu)(X - \mu)^T] \cdot u \end{aligned}$$

Define the covariance matrix:

$$\Sigma = E[(X - \mu)(X - \mu)^T].$$

Therefore: $Var(u^T X) = u^T \Sigma u$